Rajit Sharma

732-912-9895 | rajit.sharma@proton.me | linkedin.com/in/rajit00101 | github.com/rajit00101

Education

Boston University — Boston, MA
M.S. in Computer Science & Artificial Intelligence — *Expected May 2028*

Rutgers University — New Brunswick, NJ
B.S. in Finance, Second Major in Computer Science, Minor in Japanese — *Graduated May 2026*
Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Software Methodology, Principles of Information & Data Management, ML Principles

Professional Experience

Amtrak – Software Engineer — Nov. 2024 – Present
Internship — *Philadelphia, PA*

- Automated EEPS employee record workflows in **Python** and **Access** by parsing structured CSV inputs, identifying duplicate records, and generating sorted email outputs; reduced processing time from hours to minutes.
- Refactoring EEPS from a standalone script into a maintainable **Python** package using **pandas**, **pyproject.toml**, and **pytest** to improve testability and cross-machine setup.
- Debugged defects in **RDT** (Remote Dispatch Terminal), a SQL-backed client-server application using **React, JavaScript, Node.js, HTML, and CSS**, including a browser caching issue causing stale status indicators.
- Supported release validation through regression testing, Jira defect documentation, AMTEC traceability matrices, and **Confluence** documentation to improve future test coverage.

New York Life – Software Developer — June – August 2023
Internship — *New York, NY*

- Streamlined data analysis with **Python, Pandas, Jupyter Notebook, and Excel**, reducing task time from two weeks to one click with a 90% efficiency gain.
- Independently built a Python tool to tally votes from Excel input schemas and authored documentation to improve adoption and maintainability.

Projects

Employee Records Automation Tool | *Python, pandas, pytest, pyproject.toml, sqlite* — Current

- Refactoring a CSV-based employee record automation script into a modular Python project with reusable components, DataFrame-based transformations, validation logic, and automated tests.
- Improving structured data handling by replacing manual CSV generation with **pandas** DataFrame operations for duplicate detection, sorting, and formatted email output generation and **sqlite** database to validate entries.

Personal Infrastructure Lab | *Proxmox, Docker, LXC, DNS, Backups* — Fall 2025 – Current

- Design and manage a homelab environment using **Proxmox VMs, Docker containers, and LXC containers** to simulate infrastructure, service deployment, and access-control workflows.
- Deploy and maintain self-hosted services for image backup, file hosting, and custom DNS while testing backup strategies, service reliability, and fault recovery.

Pizza Factory | *Java, Gradle, FXML, Git, Bitbucket* — Fall 2024

- Built a cross-platform pizza ordering app using **Java** and **Gradle**, applying object-oriented design principles, modular UI components, and persistent data storage.
- Collaborated through **Git** and **Bitbucket** workflows to manage source control, feature development, and team-based implementation.

Technical Skills

Languages: Java, Python, JavaScript, SQL, HTML/CSS
Frameworks & Libraries: React, Node.js, pandas, pytest, Jupyter Notebook
Databases & Data: PostgreSQL, SQL, CSV processing, data validation, structured data workflows
Developer Tools: Git, Bitbucket, Docker, Jira, Confluence, pyproject.toml, VS Code, IntelliJ, PyCharm, CLI
Systems & Infrastructure: Proxmox, Docker containers, LXC containers, DNS, backups, access control
AI-Assisted Development: Claude, ChatGPT for debugging support, code comprehension, documentation, and refactoring assistance